

Name: _____

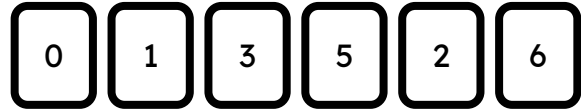
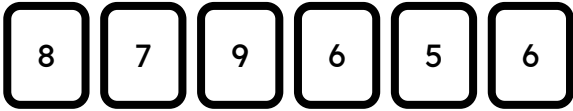


Securing understanding of place value in integers

Task A: Recognising place value

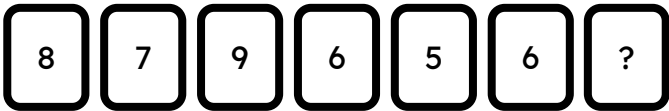
Below are two separate sets of digit cards.

For each set, use the digit cards to create:



- 1) the largest integer
- 2) the smallest integer using all cards
- 3) the integer closest to 600
- 4) the integer closest to 1000

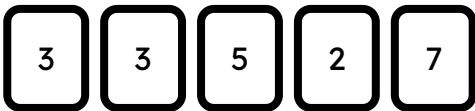
Another digit card is added to each set at random:



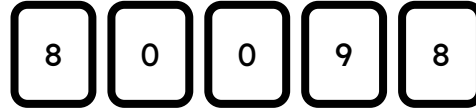
5) Is there a scenario where the second set of digits could make a larger integer than the first? Why?

6) Jun and Lucas are trying to make the smallest values possible using all of their digits cards.

Identify and correct any mistakes in their statements below.



Jun



Lucas

Jun: "The smallest number I can make is 23357"

Lucas: "The smallest number I can make is 8089"

Jun: "If I swap my 2 for one of Lucas' 0 cards I can then make the smallest number."

Task B: Integers as sums of their digits' place value

Fill in the blanks:

$$\begin{aligned} 1) \ 2372 &= 2000 + \underline{\hspace{2cm}} + 70 + 2 \\ &= 2 \times \underline{\hspace{2cm}} + 3 \times 100 + 7 \times 10 + 2 \times 1 \end{aligned}$$

$$\begin{aligned} 2) \ 36\ 220 &= \underline{\hspace{2cm}} + 6000 + 200 + 20 \\ &= 3 \times \underline{\hspace{2cm}} + 6 \times 1000 + 2 \times 100 + \underline{\hspace{2cm}} \times 10 \end{aligned}$$

$$\begin{aligned} 3) \ 8 _ 8 &= 800 + \underline{\hspace{2cm}} + 8 \\ &= \underline{\hspace{2cm}} \times 100 + 5 \times 10 + 8 \times \underline{\hspace{2cm}} \end{aligned}$$

$$\begin{aligned} 4) \ 5\ 003\ 009 &= 9 + 5\ 000\ 000 + \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \times 1 + 3 \times \underline{\hspace{2cm}} + 5 \times 1\ 000\ 000 \end{aligned}$$

$$\begin{aligned} 5) \ 17 _ 06 &= 7000 + 6 + 10\ 000 + \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \times 1000 + 6 \times 1 + 6 \times 100 + 1 \times 10\ 000 \end{aligned}$$

$$\begin{aligned} 6) \ 300 _ 80 &= \underline{\hspace{2cm}} + 200 + 300\ 000 \\ &= 8 \times \underline{\hspace{2cm}} + \underline{\hspace{2cm}} \times 100 + 3 \times 100\ 000 \end{aligned}$$

Task C: Rearranging digits in integers

1) Write down all the unique 3-digit integers you can make with the digits below.

2	1	9
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2) Choose three digits which can make exactly three unique 3-digit integers.

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3) Choose three digits which can make exactly one unique 3-digit integer.

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4) Write all the unique 4-digit integers which can be made with the digits below.

5	6	6	6
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5) Aisha arranges the digits above to make the largest integer possible. She then takes a zero digit card. Where can she place the zero card so that:

a) her new integer is the largest 5-digit integer possible

b) her new integer is as close to 61 000 as possible

c) her new integer is as close to 66 000 as possible